

Introduction to Stochastic Schrödinger Equation

Zhu Quanhui

National University of Singapore

Sept. 05, 2025

Group Meeting

Outline

- 1 Stochastic Schrödinger Equation
- 2 Derivation of the Diffusion Markov SSE
- 3 Non-Markovian Stochastic Schrödinger Equation
- 4 Numerical Examples
- 5 Appendix

Schrödinger Equation

The time-dependent Schrödinger equation for a closed quantum system is

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle,$$

where H is the system Hamiltonian and $|\psi(t)\rangle$ is the state vector.

- Describes unitary evolution of pure states.
- No interaction with environment (closed system).

For simplicity, we set $\hbar = 1$ in the following.

Open Quantum Systems

Realistic quantum systems interact with their environment, leading to:

- Decoherence: loss of quantum coherence.
- Dissipation: energy exchange with the environment.
- Non-unitary dynamics: cannot be described by Schrödinger equation alone.

The total Hamiltonian is

$$H_{\text{total}} = H_{\text{system}} \otimes \mathbb{I}_E + H_{\text{environment}} \otimes \mathbb{I}_S + H_{\text{interaction}}.$$

It leads to two main approaches:

- Master equations (ME) for the density matrix ρ .
- Stochastic Schrödinger equations (SSE) for pure-state trajectories ψ .

Lindblad Master Equation

Under Born-Markov and secular approximations, the system density matrix $\rho(t)$ obeys

$$\frac{d\rho}{dt} = -i[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right).$$

- H : effective system Hamiltonian.
- L_k : Lindblad operators (dissipation/measurement channels).
- Assumptions: short bath correlation time, no memory effects.
- Deterministic equation for the ensemble state.

Markovian Stochastic Schrödinger Equation

A stochastic pure-state evolution reproducing the same physics:

$$d|\psi_t\rangle = \left(-iH - \frac{1}{2} \sum_k L_k^\dagger L_k \right) |\psi_t\rangle dt + \sum_k L_k |\psi_t\rangle dW_t^k,$$

where dW_t^k are independent Wiener increments.

- Describes **conditional trajectories** (single-shot experiments).
- Noise is **white** (δ -correlated).

$$\mathbb{E}[dW_t^k] = 0, \quad \mathbb{E}[dW_t^k dW_t^l] = \delta_{kl} dt.$$

- Different unravellings (diffusive/jump) possible.

Stochastic Schrödinger Equations (SSE)

Two main types of SSE used in open quantum systems:

- **Diffusion-type SSE:** corresponds to continuous monitoring (homodyne/heterodyne detection), driven by Gaussian noise (dW_t).

$$d|\psi_t\rangle = \left[-iH - \frac{1}{2} \sum_k \left(L_k^\dagger L_k - \langle L_k + L_k^\dagger \rangle_t L_k + \frac{1}{4} \langle L_k + L_k^\dagger \rangle_t^2 \right) \right] |\psi_t\rangle dt + \sum_k \left(L_k - \frac{1}{2} \langle L_k + L_k^\dagger \rangle_t \right) |\psi_t\rangle dW_t^k.$$

- **Jump-type SSE:** corresponds to discrete quantum jumps (e.g., photon counting), driven by Poisson noise (dN_t).

$$d|\psi_t\rangle = \left(-iH - \frac{1}{2} \sum_k (L_k^\dagger L_k - \langle L_k^\dagger L_k \rangle_t) \right) |\psi_t\rangle dt + \sum_k \left(\frac{L_k}{\sqrt{\langle L_k^\dagger L_k \rangle_t}} - I \right) |\psi_t\rangle dN_t^k.$$

In both cases, the ensemble average reproduces the Lindblad master equation.

From SSE to Master Equation

Using Itô calculus, one can show that the ensemble average $\rho(t) = \mathbb{E}[|\psi_t\rangle\langle\psi_t|]$ of the stochastic pure states satisfies the Lindblad master equation. Let

$$\begin{aligned} P(t) &= |\psi_t\rangle\langle\psi_t|, \\ dP(t) &= d|\psi_t\rangle\langle\psi_t| + |\psi_t\rangle d\langle\psi_t| + d|\psi_t\rangle d\langle\psi_t| \\ &=: A + B + C. \end{aligned}$$

Recall that

$$d|\psi_t\rangle = \left(-iH - \frac{1}{2} \sum_k L_k^\dagger L_k\right) |\psi_t\rangle dt + \sum_k L_k |\psi_t\rangle dW_t^k.$$

Define $G = -iH - \frac{1}{2} \sum_k L_k^\dagger L_k$, then

$$A = GP(t)dt + \sum_k L_k P(t) dW_t^k.$$

From SSE to Master Equation

$$\begin{aligned} d\langle\psi| &= (d|\psi\rangle)^\dagger \\ &= \langle\psi|G^\dagger dt + \sum_k \langle\psi|L_k^\dagger dW_t^k, \end{aligned}$$

So

$$B = P(t)G^\dagger dt + \sum_k P(t)L_k^\dagger dW_t^k.$$

Notice that $dW_t^k dt = 0$, $(dt)^2 = 0$. The remaining term of C is

$$\begin{aligned} C &= \left(\sum_k L_k |\psi\rangle dW_t^k \right) \left(\sum_l \langle\psi| L_l^\dagger dW_t^l \right) \\ &= \sum_{k,l} L_k P(t) L_l^\dagger dW_t^k dW_t^l. \end{aligned}$$

From SSE to Master Equation

Taking the ensemble average and using $\mathbb{E}[dW_t^k dW_t^l] = \delta_{kl} dt$, $\mathbb{E}[dW_t^k] = 0$, we get

$$\begin{aligned} \mathbb{E} dP(t) &= \mathbb{E} \left(GP(t) + P(t)G^\dagger \right) dt + \sum_k \mathbb{E} \left(L_k P(t) + P(t)L_k^\dagger \right) \mathbb{E}(dW_t^k) \\ &\quad + \sum_{k,l} \mathbb{E}(L_k P(t)L_l^\dagger) \mathbb{E}(dW_t^k dW_l^k), \\ d\rho &= \left(G\rho + \rho G^\dagger + \sum_k L_k \rho L_k^\dagger \right) dt. \end{aligned}$$

Recall that $G = -iH - \frac{1}{2} \sum_k L_k^\dagger L_k$, so

$$\begin{aligned} \frac{d\rho}{dt} &= -iH\rho - \frac{1}{2} \sum_k L_k^\dagger L_k \rho + i\rho H - \frac{1}{2} \sum_k \rho L_k^\dagger L_k + \sum_k L_k \rho L_k^\dagger \\ &= -i[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right). \end{aligned}$$

This is exactly the Lindblad master equation for $\rho(t) = \mathbb{E}[|\psi_t\rangle \langle \psi_t|]$.

- 1 Stochastic Schrödinger Equation
- 2 Derivation of the Diffusion Markov SSE**
- 3 Non-Markovian Stochastic Schrödinger Equation
- 4 Numerical Examples
- 5 Appendix

Why Introduce Stochastic Terms?

- In practice, detectors yield **individual experimental records** (quantum trajectories).
- **Stochasticity originates from:**
 - **Measurement backaction:** the act of continuous observation perturbs the system.
 - **Environmental fluctuations:** reservoir modes act as quantum noise sources.
- Noise structure reflects measurement type:
 - Gaussian increments (dW_t) $\rightarrow$ continuous weak measurements.
 - Poisson increments (dN_t) $\rightarrow$ discrete quantum jumps.

Measurement Model: Continuous Weak Measurement

Consider a system weakly coupled to an environment (detector).

- At each step δt , the measurement is described by operators:

$$\Omega_r = \left(\frac{k\delta t}{\pi} \right)^{1/4} \exp\left(-\frac{k\delta t}{2}(r - X)^2 \right),$$

where X is the measured observable, r is the measurement result, and k characterizes measurement strength.

- Update rule:

$$|\psi(t + \delta t)\rangle = \frac{\Omega_r |\psi(t)\rangle}{\sqrt{\langle \psi | \Omega_r^\dagger \Omega_r | \psi \rangle}},$$

where $P(r) = \langle \psi | \Omega_r^\dagger \Omega_r | \psi \rangle$ is the probability density for result r .

- Completeness: $\int_{-\infty}^{\infty} \Omega_r^\dagger \Omega_r = I$.

Random Measurement Outcomes

The measurement result r is a random variable with distribution

$$r = \langle X \rangle_t + \frac{\Delta W}{\sqrt{2k\delta t}},$$

where $\Delta W \sim \mathcal{N}(0, \delta t)$ is a Wiener increment.

- Mean: $\mathbb{E}[\Delta W] = 0$.
- Variance: $\text{Var}(\Delta W) = \delta t$.
- In the continuous limit $\delta t \rightarrow 0$, $r(t)$ becomes a noisy signal:

$$r(t) = \langle X \rangle_t + \xi(t), \quad \xi(t) = \frac{dW_t}{\sqrt{2kdt}}.$$

This is a **continuous weak measurement**.

Expanding the Measurement Operator

Plug r into Ω_r :

$$\begin{aligned}\Omega_r &\propto \exp\left(-\frac{k\delta t}{2}(r - X)^2\right) \\ &= \exp\left(-\frac{k\delta t}{2}\left((X - \langle X \rangle_t) - \frac{\Delta W}{\sqrt{2k\delta t}}\right)^2\right) \\ &= \exp\left(-\frac{k\delta t}{2}(X - \langle X \rangle_t)^2 + \sqrt{\frac{k}{2}}(\langle X \rangle_t - X)\Delta W - \frac{(\Delta W)^2}{4\delta t}\right).\end{aligned}$$

In itô rule, $(\Delta W)^2 = \delta t$, and denote $\hat{X} = X - \langle X \rangle_t$. Expand Ω_r to first order in δt yields

$$\begin{aligned}\Omega_r &\propto I - \frac{k}{2}\hat{X}^2\delta t + \sqrt{\frac{k}{2}}\hat{X}\Delta W + \frac{k}{4}\hat{X}^2(\Delta W)^2 \\ &= I - \frac{k}{4}\hat{X}^2\delta t + \sqrt{\frac{k}{2}}\hat{X}\Delta W.\end{aligned}$$

Remark: **Scalar** factors do not affect the state update.

Unnormalized State Evolution

Define unnormalized state $|\tilde{\psi}_t\rangle$. In the continuous limit $\delta t \rightarrow dt$, the update is

$$|\tilde{\psi}_{t+dt}\rangle \approx \left(1 - \frac{k}{4}Xdt - \sqrt{\frac{k}{2}}XdW_t\right)|\tilde{\psi}_t\rangle.$$

When consider the Hamiltonian H , the evolution becomes

$$d|\tilde{\psi}_t\rangle = \left(-iH - \frac{k}{4}X^2\right)|\tilde{\psi}_t\rangle dt + \sqrt{\frac{k}{2}}X|\tilde{\psi}_t\rangle dW_t.$$

It is the **linear** stochastic differential equation.

Normalization and Final Form

- Redefine Lindblad operator L , include Hamiltonian H .
- In Itô form, the **linear SSE** is:

$$d|\tilde{\psi}_t\rangle = \left(-iH - \frac{1}{2} \sum_k L_k^\dagger L_k \right) |\tilde{\psi}_t\rangle dt + \sum_k L_k |\tilde{\psi}_t\rangle dW_t^k.$$

- Physical (normalized) state:

$$|\psi_t\rangle = \frac{|\tilde{\psi}_t\rangle}{\|\tilde{\psi}_t\|}.$$

From Linear to Nonlinear SSE

Let $N_t = \langle \tilde{\psi}_t | \tilde{\psi}_t \rangle$, by Itô's differential rule,

$$dN_t = \langle \tilde{\psi}_t | d\tilde{\psi}_t \rangle + \langle d\tilde{\psi}_t | \tilde{\psi}_t \rangle + \langle d\tilde{\psi}_t | d\tilde{\psi}_t \rangle.$$

Denote $G = -iH - \frac{1}{2} \sum_k L_k^\dagger L_k$, then

$$\langle \tilde{\psi}_t | d\tilde{\psi}_t \rangle = \langle \tilde{\psi}_t | G | \tilde{\psi}_t \rangle dt + \sum_k \langle \tilde{\psi}_t | L_k | \tilde{\psi}_t \rangle dW_t^k,$$

$$\langle d\tilde{\psi}_t | \tilde{\psi}_t \rangle = \langle \tilde{\psi}_t | G^\dagger | \tilde{\psi}_t \rangle dt + \sum_k \langle \tilde{\psi}_t | L_k^\dagger | \tilde{\psi}_t \rangle dW_t^k,$$

$$\langle d\tilde{\psi}_t | d\tilde{\psi}_t \rangle = \sum_{k,l} \langle \tilde{\psi}_t | L_k^\dagger L_l | \tilde{\psi}_t \rangle dW_t^k dW_t^l = \sum_k \langle \tilde{\psi}_t | L_k^\dagger L_k | \tilde{\psi}_t \rangle dt.$$

Thus

$$\begin{aligned} dN_t &= \left(\langle \tilde{\psi}_t | G + G^\dagger | \tilde{\psi}_t \rangle + \sum_k \langle \tilde{\psi}_t | L_k^\dagger L_k | \tilde{\psi}_t \rangle \right) dt + \sum_k \left(\langle \tilde{\psi}_t | L_k + L_k^\dagger | \tilde{\psi}_t \rangle \right) dW_t^k \\ &= N_t \sum_k \langle L_k + L_k^\dagger \rangle_t dW_t^k. \end{aligned}$$

From Linear to Nonlinear SSE

Using Itô's rule for $|\psi_t\rangle = \frac{|\tilde{\psi}_t\rangle}{\sqrt{N_t}}$ and dropping the high order terms, we have

$$d|\psi_t\rangle = \frac{d|\tilde{\psi}_t\rangle}{\sqrt{N_t}} - \frac{1}{2} \frac{|\tilde{\psi}_t\rangle}{\sqrt{N_t^3}} dN_t + \frac{3}{8} \frac{|\tilde{\psi}_t\rangle}{\sqrt{N_t^5}} (dN_t)^2 + d\tilde{\psi} d\left(\frac{1}{\sqrt{N_t}}\right)$$

$$\frac{d|\tilde{\psi}_t\rangle}{\sqrt{N_t}} = G|\psi_t\rangle dt + \sum_k L_k |\psi_t\rangle dW_t^k$$

$$-\frac{1}{2} \frac{|\tilde{\psi}_t\rangle}{\sqrt{N_t^3}} dN_t = -\frac{1}{2} \sum_k \langle L_k + L_k^\dagger \rangle_t |\psi_t\rangle dW_t^k$$

$$\frac{3}{8} \frac{|\tilde{\psi}_t\rangle}{\sqrt{N_t^5}} (dN_t)^2 = \frac{3}{8} \sum_k \langle L_k + L_k^\dagger \rangle_t^2 |\psi_t\rangle dt$$

$$d\tilde{\psi} d\left(\frac{1}{\sqrt{N_t}}\right) = -\frac{1}{2} \sum_k \langle L_k + L_k^\dagger \rangle_t L_k |\psi_t\rangle dt.$$

From Linear to Nonlinear SSE

Denote $m_k(t) = \langle L_k + L_k^\dagger \rangle_t$, then

$$d|\psi_t\rangle = G|\psi_t\rangle dt + \sum_k \left(\frac{3}{8} m_k(t)^2 - \frac{1}{2} m_k(t) L_k \right) |\psi_t\rangle dt + \sum_k \left(L_k - \frac{1}{2} m_k(t) \right) |\psi_t\rangle dW_t^k.$$

Check that

$$d\langle\psi_t|\psi_t\rangle = \langle\psi_t|d\psi_t\rangle + \langle d\psi_t|\psi_t\rangle + \langle d\psi_t|d\psi_t\rangle = 0.$$

$$\text{Coefficient of } dW_t^k : \langle\psi_t|(L_k - \frac{1}{2}m_k(t))|\psi_t\rangle + \langle\psi_t|(L_k^\dagger - \frac{1}{2}m_k(t))|\psi_t\rangle = 0,$$

$$\begin{aligned} \text{Coefficient of } dt : & \langle\psi_t|G + G^\dagger|\psi_t\rangle + \sum_k \langle\psi_t|(L_k^\dagger - \frac{1}{2}m_k(t))(L_k - \frac{1}{2}m_k(t))|\psi_t\rangle \\ &= \sum_k \left(-\langle L_k^\dagger L_k \rangle_t + \frac{1}{4} m_k^2 + \langle (L_k^\dagger - \frac{1}{2}m_k)(L_k - \frac{1}{2}m_k) \rangle_t \right) \\ &= 0. \end{aligned}$$

Innovation Wiener Process

Let $\hat{W}_t^k = W_t^k - \int_0^t m_k(s)ds$ be the diffusion innovation process, then $d\hat{W}_t^k = dW_t^k + m_k(t)dt$. The SSE can be rewritten as

$$d|\psi_t\rangle = \left(-iH - \frac{1}{2} \sum_k (L_k^\dagger L_k - m_k(t)L_k + \frac{1}{4}m_k(t)^2) \right) |\psi_t\rangle dt + \sum_k (L_k - \frac{1}{2}m_k(t)) |\psi_t\rangle d\hat{W}_t^k.$$

For the case $L^\dagger = L$ with single channel, the SSE reduces to

$$d|\psi_t\rangle = \left(-iH - \frac{1}{2}(L - \langle L \rangle_t)^2 \right) |\psi_t\rangle dt + (L - \langle L \rangle_t) |\psi_t\rangle dW_t.$$

- 1 Stochastic Schrödinger Equation
- 2 Derivation of the Diffusion Markov SSE
- 3 Non-Markovian Stochastic Schrödinger Equation**
- 4 Numerical Examples
- 5 Appendix

Key Formalisms

Non-Markovian Quantum State Diffusion (NMQSD/NMSSE)

[Diósi, Gisin, and Strunz, Phys. Rev., 1998.]

$$\frac{d}{dt}\psi_t = -iH\psi_t + L\psi_t z_t - L^\dagger \int_0^t \alpha(t-s) \frac{\delta\psi_t}{\delta z_s} ds,$$

$\alpha(t, s)$: bath correlation function;

z_t : colored process with $\mathbb{E}[z_t^* z_s] = \alpha(t-s)$, $\mathbb{E}[z_t z_s] = 0$.

- Recovers Markovian SSE when $\alpha(t-s) \rightarrow \delta(t-s)$.
- Long correlation time effects and memory captured by $\alpha(t, s)$.
- Complements non-Markovian master equations.

Methods for NMSSE

- **Perturbative expansions** [Gaspard, Nagaoka, JCP¹, 1999; Gambetta, Wiseman, PRA², 2003; Ke, Zhao, JCP, 2017]
 - Pros Efficient, accurate at weak coupling.
 - Cons Fails for strong environments.
- **Hierarchical methods** [Suess, Eisfeld, Strunz, PRL, 2014; Tanimura, JCP, 2020; Gao, Ren, Eisfeld, Shuai, PRA, 2022]
 - Pros Numerically exact for arbitrary coupling.
 - Cons Expensive, memory intensive.
- **Stochastic Symplectic Methods** [Chen, Hong, Jin, Sun, SINUM, 2021; Hong, Sun, 2022]
 - Pros Preserves geometric properties, stable for long-time dynamics.
 - Cons Complex implementation, moderate coupling.
- **Thermal NMSSE** [Yu, PRA, 2004; Lian, Wang, Ke, Zhao, JCP, 2019]
 - Pros Extendable to finite-temperature baths.
 - Cons Limited to weak to moderate coupling.

¹JCP: Journal of Chemical Physics

²PRA: Physical Review A, PRL: Physical Review Letters

- 1 Stochastic Schrödinger Equation
- 2 Derivation of the Diffusion Markov SSE
- 3 Non-Markovian Stochastic Schrödinger Equation
- 4 Numerical Examples**
- 5 Appendix

Two-level Amplitude Damping Model

Consider linear SSE:

$$d|\tilde{\psi}_t\rangle = \left(-iH - \frac{1}{2}L^\dagger L\right)|\tilde{\psi}_t\rangle dt + L|\tilde{\psi}_t\rangle dW_t,$$

with $H = \frac{\omega}{2}\sigma_z$, $L = \sqrt{\gamma}\sigma_-$, initial state $|\tilde{\psi}_0\rangle = |1\rangle$.

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \sigma_- = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Parameters: $\omega = 1$, $\gamma = 0.5$, $T = 5$, $\delta t = 0.001$, $n_{traj} = 400$.

The exact solution of the excited-state population is

$$p_e = \langle 1|\rho_t|1\rangle = e^{-\gamma t}.$$

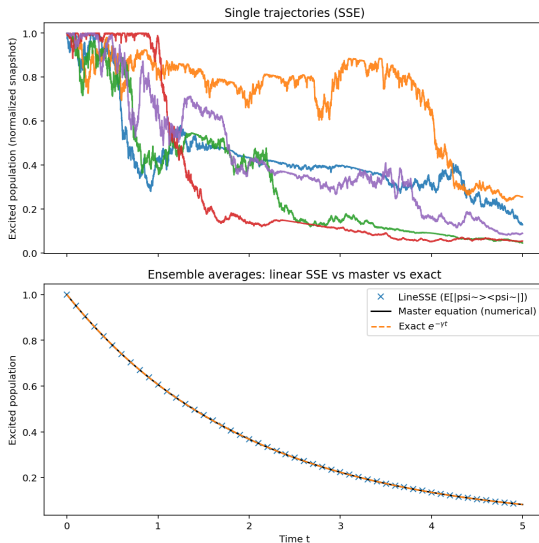


Figure: Numerical results of the linear SSE for a two-level system. The X marker denotes the average of the SSE trajectories, the black line represents the solution of the master equation, and the orange dashed line corresponds to the exact solution.

Spin Dephasing Model

In this model, $H = \frac{\omega}{2}\sigma_x$, $L = \sqrt{\gamma}\sigma_z$, initial state $|\psi_0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.

Parameters: $\omega = 1$, $\gamma = 0.2$, $T = 2$, $\delta t = 0.0002$, $n_{traj} = 50$.

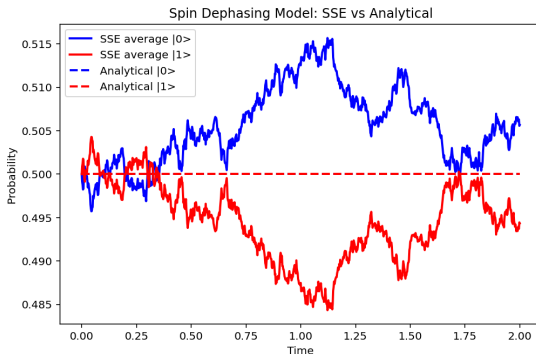


Figure: Numerical results of the SSE for a two-level system.

Appendix

- 1 Stochastic Schrödinger Equation
- 2 Derivation of the Diffusion Markov SSE
- 3 Non-Markovian Stochastic Schrödinger Equation
- 4 Numerical Examples
- 5 Appendix**

Innovation Process

Consider the normalized single-channel SSE. We define the **innovation Wiener increment** as:

$$d\widetilde{W}_t := dW_t - m_t dt,$$

where

$$m_t = \langle L + L^\dagger \rangle_t$$

is a predictable process adapted to the filtration $\mathcal{F}_t$.

Goal: prove that $\widetilde{W}_t$ is still a Wiener process.

Step 1: Martingale Property

Consider $0 \leq s < t$. The increment is

$$\widetilde{W}_t - \widetilde{W}_s = (W_t - W_s) - \int_s^t m_u du.$$

Take conditional expectation w.r.t. $\mathcal{F}_s$:

$$\mathbb{E}[\widetilde{W}_t - \widetilde{W}_s \mid \mathcal{F}_s] = \mathbb{E}[W_t - W_s \mid \mathcal{F}_s] - \mathbb{E}\left[\int_s^t m_u du \mid \mathcal{F}_s\right].$$

- Since W_t is a Wiener process: $\mathbb{E}[W_t - W_s \mid \mathcal{F}_s] = 0$. - The integral term satisfies the tower property of conditional expectation.

Step 1 (continued): Conditional Expectation of the Integral

Using the nested property:

$$\mathbb{E}\left[\int_s^t m_u du \mid \mathcal{F}_s\right] = \int_s^t \mathbb{E}[m_u \mid \mathcal{F}_s] du.$$

By definition of m_u as the predictable (adapted) part of the process:

$$\mathbb{E}[d\widetilde{W}_u \mid \mathcal{F}_u] = 0 \quad \forall u \geq s.$$

Therefore:

$$\mathbb{E}[\widetilde{W}_t - \widetilde{W}_s \mid \mathcal{F}_s] = 0,$$

which shows that $\widetilde{W}_t$ is a martingale with respect to $\mathcal{F}_t$.

Step 2: Quadratic Variation

Compute the quadratic variation of $\widetilde{W}_t$:

$$[\widetilde{W}]_t = [W - \int_0^\cdot m_s ds]_t.$$

- $[W]_t = t$ for the Wiener process. - A finite variation process $\int_0^t m_s ds$ has zero quadratic variation. - Cross term $[W, \int_0^\cdot m_s ds]_t = 0$ since one is finite variation, the other is continuous martingale.

$$\implies [\widetilde{W}]_t = t.$$

Step 3: Apply Lévy Characterization

By Lévy's characterization theorem:

If a continuous martingale M_t has quadratic variation $[M]_t = t$, then M_t is a standard Wiener process.

We have shown: - $\widetilde{W}_t$ is a continuous martingale. - $[\widetilde{W}]_t = t$.

$\implies \widetilde{W}_t$ is a Wiener process (with respect to the same filtration $\mathcal{F}_t$).

Conclusion: The innovation process is still a Wiener process.